



Stat 3113: Lesson 6

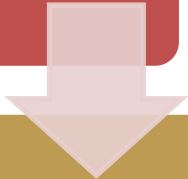
Introduction to Probability

Part 1



Learning Objectives

In this chapter, we present an **introduction** to the ideas of probability that are most important to the study of statistics.



Probability helps **quantify uncertainty** and **randomness** in data. In real-world situations, outcomes are often uncertain, and probability provides a way to model and understand this **uncertainty**.



In **statistical inference**, we often work with samples to make **inferences** about **populations**. Probability helps us analyze the distribution of statistics and make predictions about the population. Probability is the **basis for inferential statistics**, where **conclusions about populations are drawn from sample data**. Concepts such as **confidence intervals** and **hypothesis testing** rely heavily on probability distributions.

Review of Set Notation and Set Builder Notation

➤ **Set Notation:** {list items with commas in between}

- Ex: The set of all months of the year which begin with J
{January, June, July}

➤ **Set Builder Notation:** {x | describe condition for x}

The | means “such that”

- Ex: The set of solutions to the equation: $x^2 + 2x - 8 = 0$
{x | x = -4, 2}

Definitions:

Sample Space: set of all possible outcomes of an experiment.

Example 1:

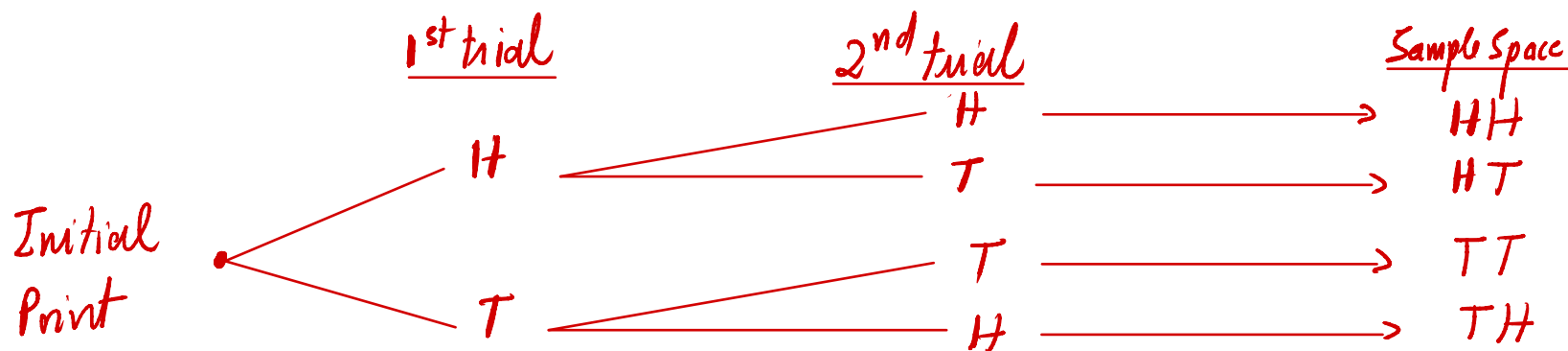
1. We toss a coin twice and record heads or tails. What is the sample space?

$$S = \{HT, HH, TH, TT\}$$

2. Roll two fair six-sided dice and record the numbers observed on each die. What is the sample space?

$$S = \{(1,1), (1,6), \dots, (6,6)\} = \{(x,y) \mid x,y \in \{1,6\}\} \text{ (Set Builder Notation)}$$

3. Let's look at a tree diagram of the experiment of tossing a coin twice.



4. We record the lifetime of a lightbulb exactly. What is the sample space?

$$\mathcal{S} = \{t \mid t \geq 0\} \quad (\text{Set Builder Notation})$$

Event: A subset of outcomes in the sample space.

Example 2: We toss a coin twice.

1. Let A be the event that we observe 2 heads.

$$A = \{HH\}$$

2. Let B be the event we observe exactly 1 head.

$$B = \{HT, TH\}$$

3. Let C be the event we observe at least 1 tail.

$$C = \{TH, HT, TT\}$$

Set Theory

Union: The union of two events A and B (denoted $A \cup B$) is all outcomes in A , in B , or in both A and B . ("or")

Intersection: The intersection of two events A and B (denoted $A \cap B$) is all outcomes in both A and B . ("and" two events both occur)

Complement: The complement of event A (denoted A^C) is all outcomes in the sample space that are not contained in A .

Example 3: (refer to Example on previous slide) Find the following sets:

1. $A \cup B = \{HH, HT, TH\} = \text{"At least 1 head"}$

2. $A \cap B = \{\} \text{ or } \emptyset$ 

Mutually Exclusive (or Disjoint): Is said of two sets if their intersection is the empty set, $\emptyset$ or $\{\}$.

a. $A^C = \{TH, HT, TT\} = C$

b. $B \cap C = \{HT, TH\} = \text{"Exactly 1 tail"}$

c. $C \cap B = B \cap C$

Example: You walk into a cafeteria and observe what the first person you see is eating for lunch. Consider the following events:

A = person had ham for lunch B = person had turkey for lunch

C = person had roast beef for lunch

What do the following mean?

a. $A \cap B = \text{Person had ham and turkey for lunch.}$

b. $C^C = \text{Person had NO roast beef for lunch}$

c. $A^C \cup B^C$ = Person had NO ham or NO turkey.

d. $A \cap B \cap C$ = Person had ham, turkey and roast beef.

e. Are A, B, and C mutually exclusive (or disjoint)? Explain.

No, because you can have ham and turkey for instance.

Example: Select one person at random.

A = person born in January

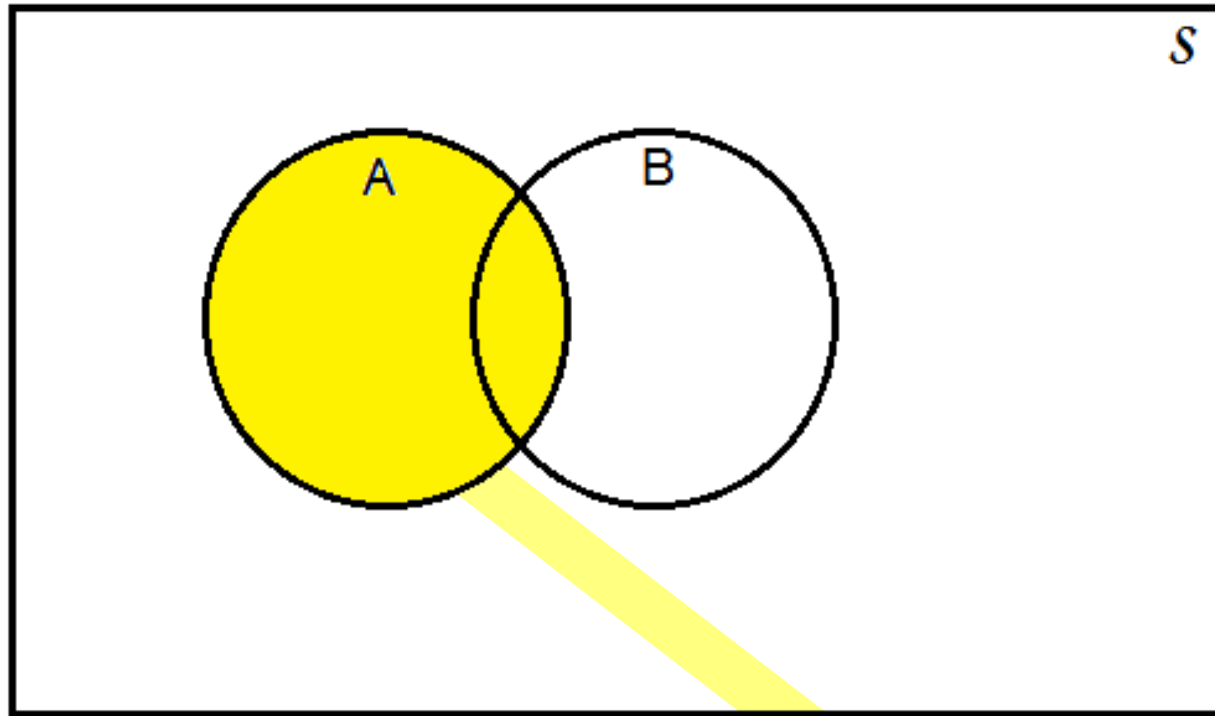
B = person born in February

C = person born in March

Etc.

Are these events mutually exclusive? Explain.

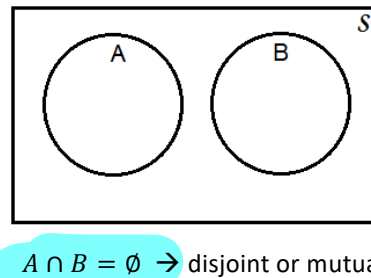
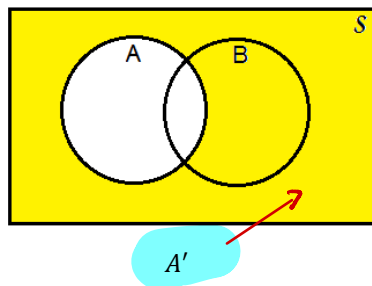
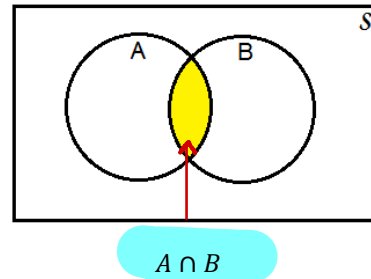
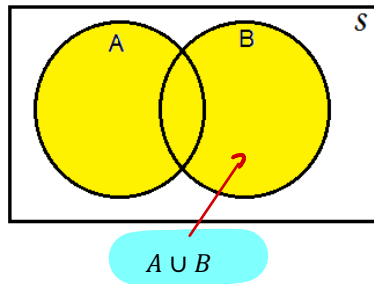
yes, because it's impossible to be born in more than a month at the same time.



Intro to Venn Diagram

- Rectangle: Inside is Sample Space
- Circles: For events (sets in sample space)
- Shading: Whatever is shaded is the set of outcomes being described
- For Example, above is a Venn Diagram for the Event A.

Common Venn Diagrams



$A \cap B = \emptyset \rightarrow$ disjoint or mutually exclusive

Reading

Chapter 3 – Section 3.1

Next Topic



Introduction to Probability **Part 2**



Chapter 3 - **Section 3.1**